

CS6P008 - Deep Learning Lab

Lab 5, Date: March 09, 2026

Topic(s): Deep Learning Optimizers

Timing: 10:00 AM to 01:00 PM

Max Points: 10

Instructions

1. Provide commented, indented code. Variables should have meaningful names.
2. Submit one .ipynb file containing all answers.
The name should be [student name]_lab[number].ipynb
3. Read the questions carefully before answering. If a question asks to follow a particular approach or to use a specific data structure, then it must be followed.
4. Write questions in separate text blocks in Jupyter/Colab Notebook before the code blocks containing answers.
5. All plots should have appropriate axis labels, titles, and legends.

Do not use the solutions provided by third-party libraries such as scikit-learn. You can use numpy, pandas, matplotlib.

1. Implement a **Multi-Layer Perceptron (MLP)** using PyTorch and train the model on the **CIFAR-10 Dataset** (Dataset Link). (7)

Tasks:

- Flatten the $32 \times 32 \times 3$ images and normalize input features.
- Implement 5 Hidden Layers with node counts: 1024, 512, 256, 128, and 64, respectively. Use **LeakyReLU** activation function for hidden layers.
- Train using **SGD**, **SGD+Momentum**, **Nesterov's Momentum**, **RMSProp** and **Adam** optimizers. **All implementations should be done in today's lab hours.**
- Compare the performance of these optimizers in the table below.
- Start with a learning rate $\eta = 0.001$, and use a learning rate scheduler such that the learning rate is reduced by a factor of 10 if the loss does not decrease for 5 consecutive epochs. Train the model until convergence.

Table 1: Performance Comparison of different Optimizers on CIFAR-10

Optimizer	Final Training Loss	Test Accuracy	Epochs to Converge
SGD			
SGD + Momentum			
Nesterov's Momentum			
RMSProp			
Adam			

2. Plot the **Epoch vs Loss** curves for all four optimizers in a single plot. Additionally, plot the **Train vs Test Accuracy** for your best-performing model. Justify which optimizer handled the high-dimensionality of CIFAR-10 best. Also, show your prediction results for a few test samples with the optimized weights of the model trained on different optimizers. (3)